Letters

RESEARCH LETTER

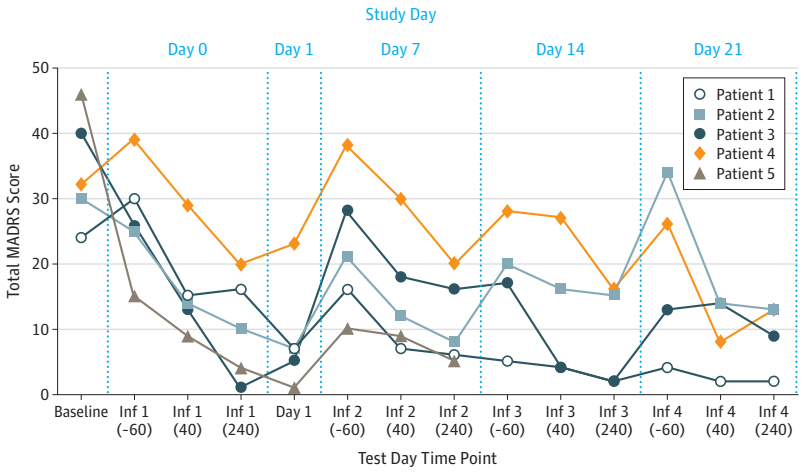
Association of Combined Naltrexone and Ketamine With Depressive Symptoms in a Case Series of Patients With Depression and Alcohol Use Disorder

Ketamine has rapid and robust antidepressant effects. However, there are concerns about the abuse liability of ketamine.[1] This concern was heightened recently owing to a preliminary report suggesting that antidepressant effects of ketamine might be dependent on opiate receptor stimulation.[2] Below, we present pilot data that indicate that the antidepressant effects of ketamine are not attenuated by naltrexone pretreatment. As a result, the combination of opiate receptor antagonism with ketamine might be a strategy to reduce addiction risk among patients with depression at risk for substance abuse.

Methods | We recruited and obtained written informed consent from 5 patients with current major depressive disorder and alcohol use disorder. In this 8-week open-label pilot study, which recieved institutional review board approval by the VA Connecticut Healthcare System Human Subjects Subcommittee, patients received injectable naltrexone (380 mg once 2-6 days prior to the first ketamine infusion) and repeated intravenous ketamine treatment (0.5 mg/kg once a week for 4 weeks; a total of 4 ketamine infusions). The study had 2 phases: (1) a 4-week ketamine treatment phase and (2) a 4-week follow-up phase. All patients were abstinent from alcohol for 5 days or longer prior to the first ketamine infusion. The primary outcome measure was clinical response defined as a 50% or higher improvement from baseline in the Montgomery-Åsberg Depression Rating Scale scores at 4 hours postinfusion.

Results | The combination of naltrexone and ketamine was associated with reduced depressive symptoms. The **Figure** shows that 60% (3 of 5) of patients met response criteria after their initial ketamine dose and 100% (5 of 5) met response criteria by their fourth dose, although 1 patient left the trial after receiving 2 ketamine infusions. The **Table** shows that depressive symptoms improved about 57% to 92%. Also, 80% (4 of 5) of patients reported improvement in alcohol craving and consumption as measured by the Obsessive Compulsive Drinking Scale. The combination treatment was safe and well tolerated in all participants. No serious adverse effects were reported in the trial.

Figure. Depressive Symptoms From Baseline to Fourth Ketamine Infusion During the Combination of Naltrexone and Ketamine Treatment



The number in parentheses indicates the minutes before or after ketamine infusion; Inf, ketamine infusion; MADRS, Montgomery-Åsberg Depression Rating Scale.

Table. Depressive Symptoms Before and After the Combination of Naltrexone and Ketamine Treatment

Patient No./Sex/Age, y	Race	MADRS Score		Improvement, No. (%)
		Pretreatment (Baseline)	Posttreatment (After Final Infusion)	
1/M/60	White	24	2	22 (92)
2/M/45	White	30	13	17 (57)
3/F/61	White	40	9	31 (78)
4/M/48	White	32	13	19 (59)
5/M/32	White	46	5	41 (89)

Abbreviations: F, female; M, male; MADRS, Montgomery-Åsberg Depression Rating Scale.

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Discussion | Our pilot data suggest that naltrexone pretreatment did not interfere with the antidepressant effects of ketamine and might enhance the treatment of comorbid alcohol use disorder. This result conflicts with that reported by Williams et al[2] in which pretreatment with 50 mg of naltrexone reduced the rate of clinical response to ketamine from 71% (5 of 7 individuals) to 0% (0 of 7 individuals). Their data and an editorial by George,[3] although preliminary, make a case for a central role for opiate agonism in the antidepressant effects of ketamine. Although our pilot data were collected under somewhat different conditions than those of Williams et al[2] (eg, different primary outcome time of 4 hours vs 1 day postinfusion, presence vs absence of alcohol use disorder, injectable vs oral naltrexone), they do not support the hypothesis that opiate receptor stimulation mediates the antidepressant effects of ketamine. Since Williams et al[2] did not provide depression ratings over a 4-hour period postinfusion, we cannot examine whether 50 mg of oral naltrexone blunted ketamine response in this early 4-hour period. Our findings are consistent with an earlier study in healthy individuals showing that the behavioral effects of an antidepressant dose of ketamine were not altered by pretreatment with 25 mg of naltrexone,[4] and some preclinical evidence that ketamine isomers may be weak partial agonists at μ opiate receptors.[5]

The initial report by Williams et al[2] and our preliminary data should be interpreted with great caution. Larger randomized clinical trials are needed to better understand whether opiate receptor stimulation contributes to the antidepressant effects of ketamine. If so, then preclinical research will be needed to help us to understand this role for opiates and its implications for future rapid-acting antidepressant treatments.

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Conflict of Interest Disclosures: Dr Petrakis reports personal fees from Alkermes outside the submitted work and has a patent to combination of opioid antagonist with ketamine pending. Dr Krystal has been a consultant for AstraZeneca Pharmaceuticals, Biogen, Biomedisyn Corporation, Bionomics, Concert Pharmaceuticals, Heptares Therapeutics, Janssen Research & Development, L.E.K. Consulting, Otsuka America Pharmaceutical, Spring Care Inc, Sunovion Pharmaceuticals, Takeda, and Taisho Pharmaceutical Co; is on the scientific advisory boards of BiOasis Technologies, Biohaven Pharmaceuticals, BlackThorn Therapeutics, Broad Institute of Massachusetts Institute of Technology and Harvard University, Cadent Therapeutics, Lohocla Research Corporation, Pfizer Pharmaceuticals, and Stanley Center for Psychiatric Research at the Broad Institute; has stock in ArRETT Neuroscience, BlackThorn Therapeutics, Biohaven Pharmaceuticals, and Spring Care; has stock options with Biohaven Pharmaceuticals; has the following patents and inventions: Seibyl JP, Krystal JH, Charney DS, Dopamine and noradrenergic reuptake inhibitors in treatment of schizophrenia, US patent number 5,447,948, September 5, 1995; Coric V, Krystal JH, Sanacora G, Glutamate modulating agents in the treatment of mental disorders, US patent number 8,778,979, July 15, 2014; Coric V, Krystal JH, Sanacora G, Glutamate agents in the treatment of mental disorders, US patent application number 15/695,164, September 5, 2017. Charney D, Krystal JH, Manji H, Matthew S, Zarate C, Intranasal administration of ketamine to treat depression, US patent number 14/197,767, March 5, 2014; Charney DS, Manji HK, Krystal JH, Matthew SJ, Zarate CA, Intranasal administration of ketamine to treat depression, US application or Patent Cooperation Treaty International application number 14/306,382, filed on June 17, 2014; Zarate C, Charney DS, Manji HK, Mathew, Sanjay J, Krystal JH, Methods for treating suicidal ideation, US patent application number 14/197,767 filed on March 5, 2014, by Yale University; Arias A, Petrakis I, Krystal JH, Composition and methods to treat addiction, provisional use patent application number 61/973/961 filed on April 2, 2014, by Yale University; Chekroud A, Gueorguieva R, Krystal JH, Treatment selection for major depressive disorder, June 3, 2016, USPTO docket number Y0087.70116US00, provisional patent submission by Yale University; and Abdallah, C, Krystal, JH, Duman, R, Sanacora, G. Combination Therapy for Treating or Preventing Depression or Other Mood Diseases. USPTO No. 047162-7177P1 (00754) filed on August 20, 2018 by Yale University; and has received nonfederal research support from AstraZeneca Pharmaceuticals, which provides the drug saracatinib for research related to US National Institute on Alcohol Abuse and Alcoholism grant Center for Translational Neuroscience of Alcoholism, and from Pfizer Pharmaceuticals, which provides an investigational drug, PF-03463275, for research related to US National Institutes of Health grant Translational Neuroscience Optimization of GlyT1 Inhibitor. All authors (Drs Yoon, Petrakis, and Krystal) are listed as inventors on a patent application by Yale University (application no. PCT/US2017/056922; Compounds, Compositions and Methods for Treating or Preventing Depression and Other Diseases).

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